

ELASTOLAB: A Physics-Informed Computational Framework for Ultrasound Elastography, Tissue Stiffness Reconstruction, and Uncertainty-Aware Diagnosis

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Abstract

Ultrasound elastography provides a non-invasive approach for characterizing tissue mechanical properties through tissue deformation and shear-wave propagation, complementing conventional ultrasound with quantitative stiffness information [1], [2], [3]. However, accurate stiffness characterization remains challenging because of heterogeneous tissue properties, sparse observations, complex wave propagation, and the inverse nature of elastography reconstruction [4], [5]. In this work, we present ELASTOLAB, a virtual ultrasound elastography laboratory that integrates tissue mechanics, numerical wave propagation, synthetic data generation, machine learning, physics-informed neural networks (PINNs), and uncertainty-aware prediction within a unified computational framework. The physics layer models tissue stiffness through Young's modulus, shear-wave velocity, and strain relationships, while a two-dimensional finite-difference time-domain (FDTD) solver simulates wave propagation through heterogeneous tissue phantoms [6], [7], [8], [9]. These simulations generate synthetic elastography samples containing B-mode representations, strain maps, ground-truth stiffness maps, and tissue labels. Quantitative elastography features are evaluated using classical machine-learning models and a dual-channel convolutional neural network combining B-mode and strain information, building on established machine-learning approaches for breast elastography [10], [11], [12]. A central component of ELASTOLAB is a PINN for reconstructing spatial stiffness distributions from sparse observations. The network incorporates data, physics, and boundary-condition constraints following the established PINN framework for forward and inverse problems [13]. Recent studies have further demonstrated the application of physics-informed learning to elasticity imaging, tissue-property reconstruction, and shear-wave-based viscoelastic characterization [14], [15], [16], [17], [18]. The framework additionally employs conformal prediction to quantify predictive uncertainty and assess the reliability of model decisions [19]. Model performance is evaluated using classification, reconstruction, statistical, and ablation analyses. Overall, ELASTOLAB provides an end-to-end and reproducible computational environment for investigating the relationship between ultrasound observations, tissue mechanics, wave propagation, tissue stiffness, and machine-learning-based interpretation. The framework is designed to support controlled experimentation on physics-informed and uncertainty-aware elastography and to provide a foundation for future validation using real ultrasound datasets such as BUSI [20].

Keywords: Ultrasound elastography, tissue stiffness reconstruction, finite-difference time-domain (FDTD), physics-informed neural networks (PINNs), machine learning, shear-wave elastography, uncertainty quantification, conformal prediction.

1. INTRODUCTION

Ultrasound is one of the most widely used medical imaging modalities because of its non-invasive nature, real-time capability,

relatively low cost, and absence of ionizing radiation. Conventional B-mode ultrasound primarily provides anatomical information based on acoustic intensity and tissue interfaces. However, many pathological processes are also

associated with changes in the mechanical properties of tissue. Ultrasound elastography extends conventional ultrasound by providing information about tissue elasticity, enabling mechanical properties to be assessed through tissue deformation or shear-wave propagation [1], [2], [3].

Tissue stiffness is clinically relevant in several applications, particularly in the characterization of breast lesions and other soft-tissue abnormalities. Strain elastography estimates tissue deformation under mechanical or physiological excitation, whereas shear-wave elastography estimates mechanical properties from the propagation of mechanically induced shear waves [4], [21]. Quantitative elastography has demonstrated the ability to distinguish tissues with different mechanical characteristics, including benign and malignant breast lesions [22], [23], [24]. Clinical interpretation has also incorporated qualitative and semi-quantitative approaches, including elastographic pattern classification and scoring systems [11], [25]. Acoustic radiation force impulse imaging provides another important approach for generating localized tissue displacement and assessing mechanical properties [26].

Despite these capabilities, computational analysis of elastography remains challenging. Tissue is mechanically heterogeneous, wave propagation depends on spatially varying material properties, and clinical measurements may be sparse or affected by noise and acquisition conditions. Consequently, reconstructing a complete stiffness distribution from limited observations constitutes an inverse problem. Numerical simulation provides a controlled environment for studying these relationships by allowing tissue properties and excitation conditions to be specified explicitly. Finite-difference approaches have been used to model acoustic and mechanical wave propagation in heterogeneous and viscoelastic media, including applications related to medical elastography [7], [8], [9], [27]. Computational ultrasound frameworks such as k-Wave further demonstrate the value of numerical wave-field simulation for medical imaging research [6].

Machine learning provides an alternative approach for extracting complex patterns from

elastography data. Deep-learning methods have increasingly been investigated for ultrasound elastography analysis, including breast-lesion classification using shear-wave elastography images [5], [10], [12]. However, purely data-driven models can require substantial training data and may produce predictions that are difficult to relate to the underlying physical behavior of tissue. This motivates the integration of physical knowledge directly into learning-based models.

Physics-informed neural networks (PINNs) provide a framework for incorporating governing physical relationships into neural-network optimization by adding physics-based constraints to the learning objective [13]. Recent studies have applied physics-informed learning to elasticity reconstruction and the estimation of tissue mechanical properties, demonstrating its potential for inverse elasticity problems [14], [15], [16], [17]. More recent work has also explored physics-informed deep learning for shear-wave viscoelasticity imaging [18]. These developments suggest that combining numerical simulation with physics-informed learning can provide a promising direction for reconstructing tissue properties from limited observations.

However, a complete computational environment that connects tissue mechanics, wave propagation, synthetic ultrasound generation, elastography feature extraction, machine-learning classification, stiffness reconstruction, and uncertainty-aware prediction remains valuable for systematic experimentation. In particular, uncertainty is important when computational models are applied to potentially diagnostic decisions. Conformal prediction provides a distribution-free framework for constructing prediction sets with a specified statistical coverage level, offering a practical mechanism for quantifying predictive uncertainty [19].

To address these challenges, this work introduces *ELASTOLAB*, a virtual ultrasound elastography laboratory designed as an integrated computational framework for studying the relationship between tissue mechanics, ultrasound observations, wave propagation, tissue stiffness, and machine learning. The

framework begins with a tissue-physics layer that relates Young’s modulus to shear-wave velocity and strain. A two-dimensional finite-difference time-domain (FDTD) simulation layer then models wave propagation through heterogeneous tissue phantoms and produces simulated ultrasound and elastography representations. These simulations form a synthetic dataset containing B-mode images, strain maps, ground-truth stiffness maps, and tissue labels. The resulting data are analyzed using quantitative elastography features, classical machine-learning models, and a dual-channel convolutional neural network.

The framework further incorporates a physics-informed neural network for spatial stiffness reconstruction from sparse observations. Unlike a purely data-driven reconstruction model, the proposed formulation incorporates data fidelity, physical consistency, and boundary-condition constraints into the learning objective. The system also includes stiffness regression, statistical analysis, reconstruction-quality assessment, and conformal prediction to evaluate both predictive performance and reliability. The framework is designed to support controlled experiments before subsequent validation on real ultrasound datasets, including publicly available breast ultrasound data such as BUSI [20].

The main contributions of this work are as follows:

- A unified virtual ultrasound elastography laboratory integrating tissue mechanics, numerical wave propagation, synthetic data generation, machine learning, physics-informed reconstruction, and uncertainty quantification.
- A two-dimensional FDTD-based simulation framework for modelling wave propagation through heterogeneous tissue phantoms and generating corresponding elastography observations.
- A synthetic elastography dataset and quantitative feature-extraction pipeline for investigating tissue stiffness and benign–malignant classification.
- A multi-model learning framework combining classical machine learning and dual-channel convolutional neural networks using complementary B-mode and strain information.
- A physics-informed neural network for reconstructing spatial stiffness distributions from sparse observations using data, physics, and boundary-condition constraints.
- An uncertainty-aware prediction layer based on conformal prediction, together with statistical testing, reconstruction metrics, model comparison, and ablation analysis for systematic evaluation.

Through this integrated design, ELASTOLAB provides a reproducible computational platform for investigating physics-informed and uncertainty-aware approaches to ultrasound elastography and establishes a foundation for future experiments using real clinical ultrasound and elastography data.

2. LITERATURE REVIEW

2.1. Ultrasound Elastography and Tissue Stiffness Characterization

Elastography was introduced as a quantitative imaging technique for estimating the mechanical properties of biological tissues by observing tissue deformation under applied stress [1]. Subsequent developments established ultrasound elastography as a practical non-invasive technique for assessing tissue stiffness. Clinical guidelines distinguish major approaches including strain-based elastography and shear-wave-based techniques, with the latter estimating tissue mechanical properties from the propagation of mechanically induced shear waves [2], [21]. Comprehensive reviews have demonstrated applications of ultrasound elastography across the breast, liver, thyroid, prostate, kidney, and other organs [3], [4].

Breast imaging has been an important application of ultrasound elastography because malignant lesions frequently exhibit different mechanical properties from surrounding tissue. Early clinical work demonstrated the diagnostic potential of ultrasound elastography for breast disease [22].

Quantitative shear-wave imaging subsequently enabled direct estimation of tissue stiffness, with studies demonstrating differences between benign and malignant breast lesions [23], [24]. Qualitative elastographic patterns have also been investigated for lesion classification [11], while dedicated clinical recommendations have been developed for breast elastography [25]. Acoustic radiation force impulse imaging further established the feasibility of generating localized tissue displacement for mechanical-property assessment [26].

Although these techniques provide clinically useful information, elastography measurements can be influenced by acquisition conditions, tissue heterogeneity, operator-dependent factors, and the assumptions used to convert measured deformation or wave propagation into quantitative mechanical properties. These limitations motivate computational approaches capable of modelling tissue mechanics and systematically investigating the relationship between observed ultrasound signals and underlying stiffness.

2.2. Numerical Simulation and Wave Propagation

Numerical simulation provides a controlled environment for studying acoustic and mechanical wave propagation in tissue. Computational methods can represent heterogeneous tissue structures with known mechanical properties and generate corresponding wave responses, enabling quantitative investigation of elastography without requiring physical measurements for every experimental condition. Finite-difference approaches have been used to simulate wave propagation in biological and tissue-like media, including heterogeneous and viscoelastic materials [7], [8], [9], [27].

The k-Wave framework demonstrated the broader utility of numerical wave-field simulation for ultrasound and photoacoustic imaging by providing computational tools for modelling acoustic propagation and reconstruction [6]. Finite-difference-based approaches are particularly attractive for a virtual elastography laboratory because spatially varying material

properties can be incorporated directly into the computational domain. This allows tissue phantoms with different stiffness regions to be constructed while controlling excitation conditions, spatial resolution, temporal sampling, and numerical stability.

However, numerical simulation alone does not solve the inverse problem of recovering tissue properties from observations. A simulated wave field can be generated from a known stiffness distribution, but reconstructing that distribution from sparse or noisy observations remains challenging. This motivates the integration of numerical simulation with learning-based inverse reconstruction.

2.3. Machine Learning for Ultrasound Elastography

Machine learning has increasingly been investigated as a means of extracting diagnostic and quantitative information from ultrasound elastography. A systematic scoping review of breast ultrasound elastography reported growing use of machine-learning methods for differentiating benign and malignant lesions and highlighted the potential of combining quantitative elastographic information with computational models [10]. More broadly, deep learning has been applied to ultrasound elastography using architectures including multilayer perceptrons, convolutional neural networks, and recurrent neural networks [5].

Convolutional neural networks are particularly suitable for elastography because both B-mode images and elastographic maps contain spatially structured information. Fujioka et al. investigated CNN-based classification of breast masses using ultrasound shear-wave elastography images, demonstrating the feasibility of learning discriminative representations directly from elastographic imaging data [12]. Such approaches reduce the need for manually designed features but remain fundamentally data-driven.

A limitation of conventional machine learning is that the model is not inherently required to obey the physical laws governing tissue mechanics or wave propagation. Consequently, a model may achieve strong predictive performance while

producing physically implausible intermediate representations or stiffness estimates, particularly when training data are limited. This limitation motivates physics-informed learning approaches.

2.4. Physics-Informed Neural Networks for Elasticity Reconstruction

Physics-informed neural networks provide a framework for incorporating governing differential equations directly into neural-network optimization. Rather than learning solely from observed input-output pairs, PINNs introduce physics-based residuals as additional components of the training objective, allowing the learned solution to satisfy known physical constraints [13].

PINNs have subsequently been investigated for elasticity-related inverse problems. Ragoza and Batmanghelich applied physics-informed neural networks to tissue elasticity reconstruction in magnetic resonance elastography, demonstrating reconstruction from simulated and in-vivo data and improved robustness to noise compared with numerical baselines [14]. Their work demonstrates that physics-informed learning can be used to solve tissue-elasticity inverse problems while incorporating physical relationships into the reconstruction process.

Physics-informed deep learning has also been applied directly to ultrasound elastography. Chen and Gu introduced a physics-informed framework capable of reconstructing the Young's modulus field of heterogeneous objects and simultaneously estimating Young's modulus and Poisson's ratio without relying on the assumption of material incompressibility [16]. Similarly, Kamali et al. investigated physics-informed elasticity imaging for spatial discovery of elastic modulus and Poisson's ratio [15]. These studies demonstrate that physics-informed learning can provide a physically constrained alternative to purely data-driven elasticity reconstruction.

More recent work has extended physics-informed learning toward shear-wave and viscoelasticity imaging. Yin et al. investigated the use of PINNs for quantitative characterization of viscoelastic properties from shear waves

across multiple organs, while Lin et al. introduced a physics-informed deep-learning approach for shear-wave viscoelasticity imaging [17], [18]. These developments demonstrate the continuing evolution of physics-informed learning from general elasticity reconstruction toward wave-based tissue-property estimation.

Despite this progress, existing studies typically focus on a particular inverse-reconstruction problem rather than providing a complete experimental environment that connects tissue modelling, wave simulation, synthetic ultrasound generation, diagnostic classification, stiffness reconstruction, uncertainty quantification, and systematic component-level evaluation.

2.5. Uncertainty Quantification and Reliable Prediction

High predictive performance alone is insufficient for computational systems intended for biomedical applications. A model should also provide information about the reliability of its predictions, particularly when observations are sparse or when the input differs from the training distribution. Conformal prediction provides a statistically grounded framework for constructing prediction sets with specified coverage without requiring strong assumptions about the underlying predictive model [19].

Within ultrasound elastography, uncertainty-aware prediction is particularly relevant because stiffness measurements and model outputs can be affected by acquisition variability, tissue heterogeneity, simulation assumptions, and limited observations. Integrating conformal prediction with machine-learning-based elastography therefore provides a mechanism for distinguishing confident predictions from cases requiring additional assessment.

2.6. Research Gap

The reviewed literature demonstrates substantial progress in four related directions: clinical ultrasound elastography for tissue characterization [3], numerical simulation of acoustic and mechanical wave propagation [6], [8], [9], machine-learning-based interpretation of elastography data [5], [12], and physics-informed

reconstruction of tissue mechanical properties [14], [15], [16], [17], [18]. However, these directions are commonly investigated as separate methodological problems.

ELASTOLAB addresses this fragmentation by combining these components into a unified virtual laboratory. The framework links tissue mechanical modelling to FDTD wave propagation, uses the resulting simulations to generate controlled elastography observations, evaluates both engineered features and image-based machine-learning models, reconstructs spatial stiffness from sparse observations using a PINN, and incorporates conformal prediction for uncertainty-aware decision support. The resulting framework is therefore positioned not as a claim of introducing PINNs or machine learning to elastography for the first time, but as an integrated computational platform for systematically studying their interaction with numerical tissue simulation, sparse-data reconstruction, and uncertainty-aware evaluation.

3. METHODOLOGY AND DATA

3.1. Overall Framework

ELASTOLAB is designed as an end-to-end virtual ultrasound elastography laboratory that connects tissue mechanics, numerical wave propagation, synthetic data generation, quantitative elastography, machine learning, stiffness reconstruction, and uncertainty-aware prediction. The framework is organized into four major layers: a physics layer, a data layer, a machine-learning layer, and a safety and evaluation layer. This organization enables controlled experimentation in which the underlying tissue properties are known during simulation and can therefore be compared directly with reconstructed or predicted properties.

The complete computational workflow is illustrated in Fig. 1. The physics layer establishes the relationship between tissue stiffness and observable mechanical quantities. The numerical simulation layer generates wave responses from heterogeneous tissue phantoms. These simulated observations are transformed into B-mode representations,

strain maps, and quantitative elastography features. The resulting data are subsequently provided to classical machine-learning models, a dual-channel convolutional neural network (CNN), and a physics-informed neural network (PINN). Finally, conformal prediction and statistical evaluation are used to assess prediction reliability and model behavior.

3.2. Tissue Physics Model

The first stage of ELASTOLAB establishes a simplified computational relationship between tissue mechanical properties and observable elastography quantities. Young’s modulus E is used as the principal representation of tissue stiffness. For an approximately incompressible soft tissue model with density ρ , the shear-wave velocity is related to Young’s modulus through

$$c_s = \sqrt{\frac{E}{3\rho}}, \quad (1)$$

which can be rearranged as

$$E = 3\rho c_s^2. \quad (2)$$

These relationships provide the physical basis for estimating stiffness from simulated shear-wave propagation. The implementation also models strain under compression and derives stiffness and strain ratios used as quantitative elastography descriptors. The tissue database contains nine tissue categories with associated mechanical-property ranges derived from published elastography literature.

The implemented tissue-physics relationships and representative stiffness parameters are summarized in Fig. 2. Ultrasound elastography fundamentally relies on the relationship between mechanical tissue properties and measurable displacement, deformation, or wave-propagation behavior [1], [2], [4], [21].

3.3. FDTD Wave Propagation Simulation

To create controlled ultrasound observations, ELASTOLAB implements a two-dimensional finite-difference time-domain (FDTD) solver. The simulation considers a spatially varying wave-speed field $c(x, y)$ representing

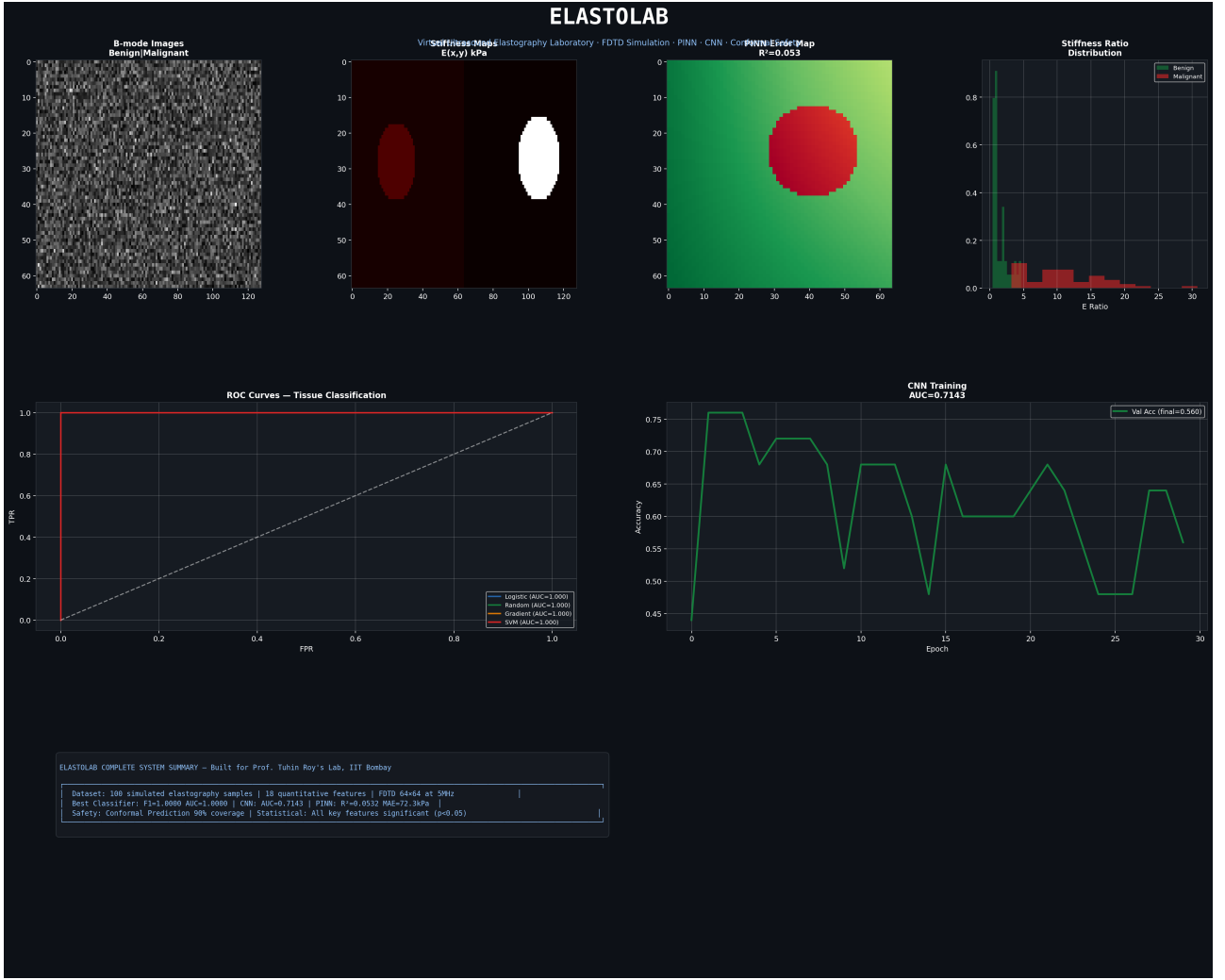


Figure 1. Overview of the ELASTOLAB computational framework. The pipeline integrates tissue mechanics, FDTD wave propagation, synthetic elastography data generation, quantitative feature extraction, machine-learning classification and regression, PINN-based stiffness reconstruction, and conformal uncertainty quantification.

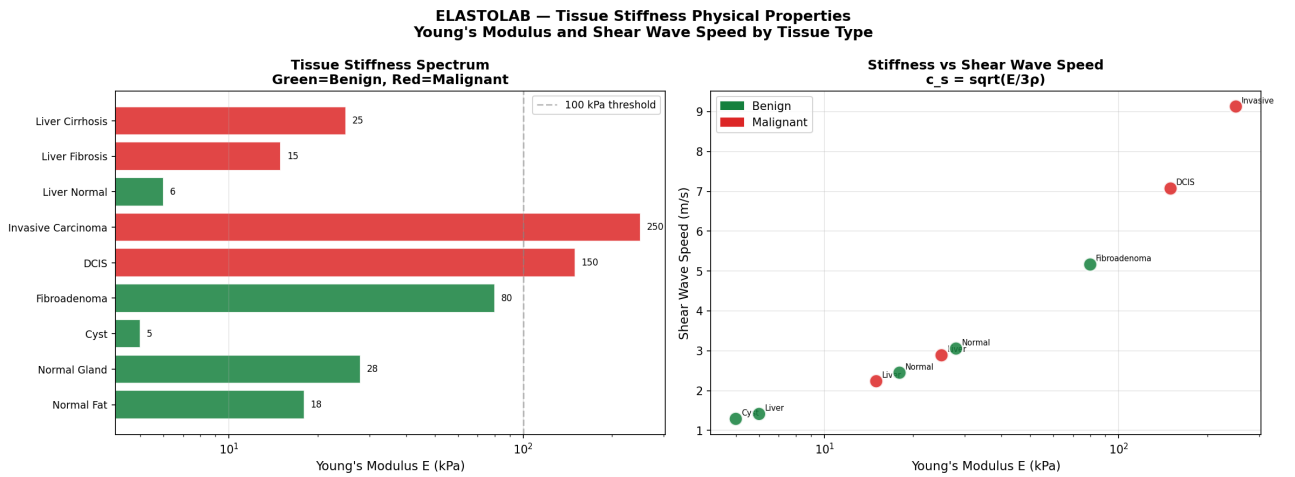


Figure 2. Tissue-physics model implemented in ELASTOLAB. The module relates Young's modulus to shear-wave velocity and strain-derived quantities and provides the mechanical-property parameters used by the subsequent simulation and reconstruction stages.

heterogeneous tissue. The scalar wave equation is expressed as

$$\frac{\partial^2 u(x, y, t)}{\partial t^2} = c^2(x, y) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad (3)$$

where $u(x, y, t)$ represents the simulated wave field and $c(x, y)$ denotes the local propagation velocity.

The spatial domain is discretized into a regular grid and the temporal evolution is calculated using finite differences. Heterogeneous breast-tissue phantoms are constructed by embedding circular inclusions with different mechanical properties within a background medium. This provides explicit ground truth for the location and stiffness of the simulated lesion. Finite-difference approaches have previously been used for modelling acoustic and mechanical wave propagation in biological and viscoelastic media [7], [8], [9], [27], [28].

Numerical stability is controlled using the Courant–Friedrichs–Lewy (CFL) condition. For the two-dimensional formulation used here, the implementation satisfies

$$\frac{c\Delta t}{\Delta x} \leq \frac{1}{\sqrt{2}}, \quad (4)$$

where Δt is the temporal step, Δx is the spatial grid spacing, and c represents the relevant propagation velocity. Computational wave-field simulation provides a controlled mechanism for investigating ultrasound propagation and image formation [6].

3.4. Shear-Wave Elastography Simulation

ELASTOLAB additionally implements a simplified shear-wave elastography (SWE) simulation to connect simulated wave propagation with quantitative stiffness estimation. Tissue regions with different stiffness values generate different shear-wave velocities according to Eq. 1. The simulated wave field is analyzed to estimate propagation velocity, which is subsequently converted into an estimated Young’s modulus using Eq. 2.

The SWE component is motivated by the physical principles underlying clinical shear-wave elastography and acoustic-radiation-force-based techniques [21], [26]. Previous computational studies have demonstrated numerical modelling of shear-wave propagation in heterogeneous and viscoelastic media [8], [9].

3.5. Synthetic Dataset Generation

Because the simulation provides direct access to the underlying tissue properties, ELASTOLAB generates a controlled synthetic dataset for evaluating the complete pipeline. The current dataset contains 100 simulated samples, consisting of 50 benign and 50 malignant cases. Each sample contains a tissue phantom, simulated FDTD response, B-mode representation, strain map, ground-truth Young’s-modulus map, and corresponding tissue label.

The synthetic design enables the learning algorithms to be evaluated against known ground-truth mechanical properties rather than relying exclusively on experimentally measured labels. It also allows tissue stiffness, lesion geometry, and observation sparsity to be systematically varied.

Figure 5 presents representative samples from the generated dataset.

3.6. Dataset Organization and Real-Data Integration

The synthetic dataset is organized to preserve the relationship between the simulated observations and their corresponding ground-truth mechanical properties. This allows B-mode images, strain maps, stiffness maps, extracted features, and labels to remain associated throughout the machine-learning and reconstruction stages.

The framework also provides an integration pathway for real publicly available datasets. In particular, the implementation includes guidance for incorporating the Breast Ultrasound Images (BUSI) dataset [20]. The dataset integration component is intended to support future experiments involving preprocessing, transfer learning, and validation using real ultrasound observations. The present study primarily evaluates the computational framework using controlled synthetic data, while real-data integration is provided as a pathway for subsequent validation.

Figure 6 summarizes the dataset organization and integration workflow.

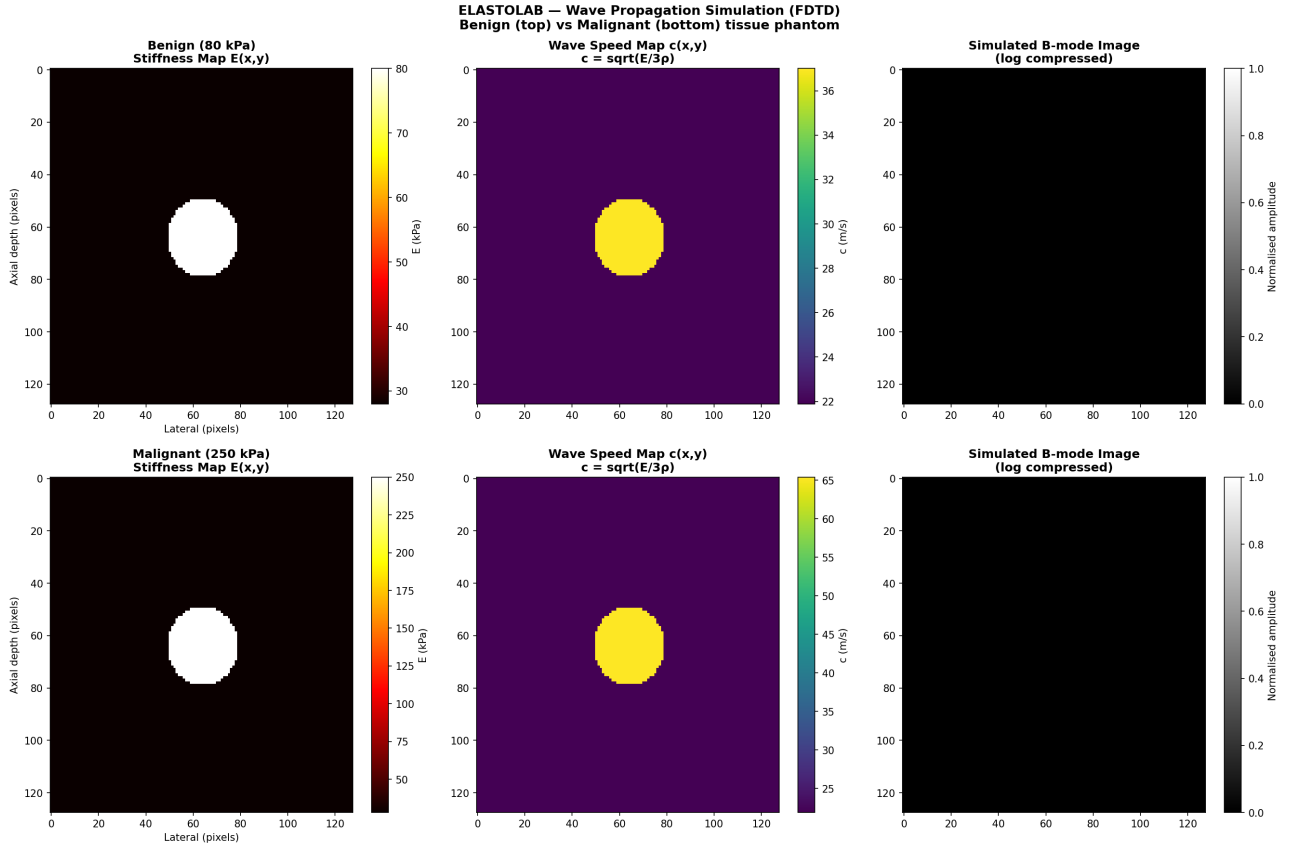


Figure 3. FDTD wave-propagation simulation in a heterogeneous tissue phantom. The simulation models propagation through a background tissue medium containing a circular inclusion with different local wave speed and stiffness.

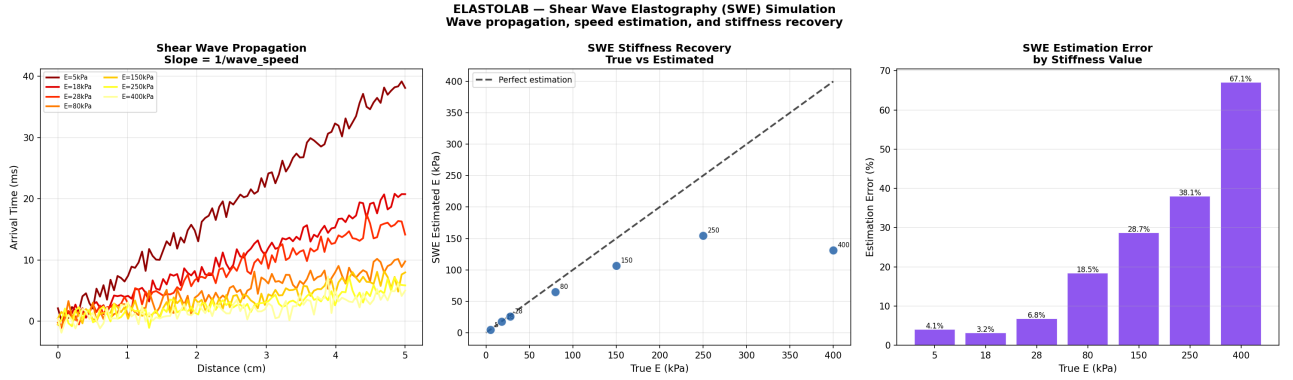


Figure 4. Simulated shear-wave elastography analysis. Wave propagation is evaluated across tissue regions with different stiffness values, allowing shear-wave velocity to be estimated and mapped to tissue stiffness.

3.7. Quantitative Elastography Feature Extraction

To provide interpretable numerical descriptors for machine learning, ELASTOLAB extracts 18 quantitative features from the simulated elastography data. These features characterize tissue stiffness, deformation, lesion boundaries, image contrast, variability, and texture.

Representative features include stiffness ratio,

strain ratio, boundary sharpness, coefficient of variation, contrast-to-noise ratio (CNR), and an approximated Tsukuba elastographic score. The stiffness and strain ratios quantify differences between the simulated lesion and surrounding reference tissue, while boundary and texture measures characterize spatial and intensity differences.

Clinical elastography studies have established the relevance of quantitative stiffness and strain

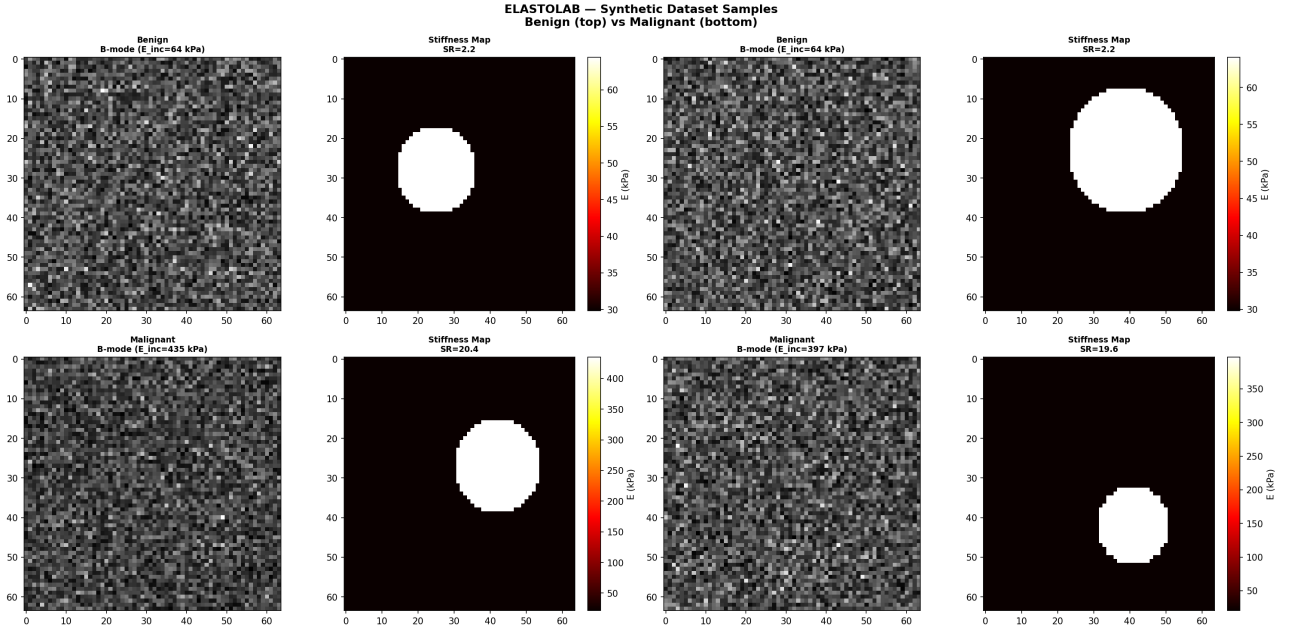


Figure 5. Representative samples from the ELASTOLAB synthetic elastography dataset. Each simulated case contains corresponding imaging, elastographic, stiffness, and tissue-label information generated from the underlying tissue phantom.



Figure 6. Dataset organization and real-data integration workflow in ELASTOLAB. The framework maintains corresponding simulated observations and ground-truth mechanical properties while providing an interface for future integration of publicly available ultrasound datasets.

measurements for breast-lesion characterization [22], [23], [24]. The use of elastographic patterns and scoring has also been investigated for differentiating benign and malignant lesions [11], [25].

3.8. Classical Machine-Learning Classification

The extracted quantitative features are used to train and evaluate four classical classification algorithms: Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), and Support Vector Machine (SVM). These models

provide complementary baselines for evaluating whether quantitative elastography measurements contain sufficient information to distinguish the simulated benign and malignant classes.

Model evaluation includes cross-validation, receiver operating characteristic (ROC) analysis, area under the ROC curve (AUC), and permutation-based feature importance. This enables both predictive performance and the relative contribution of individual elastography features to be examined. Machine-learning-based analysis of ultrasound elastography has previously been investigated for breast-tumour

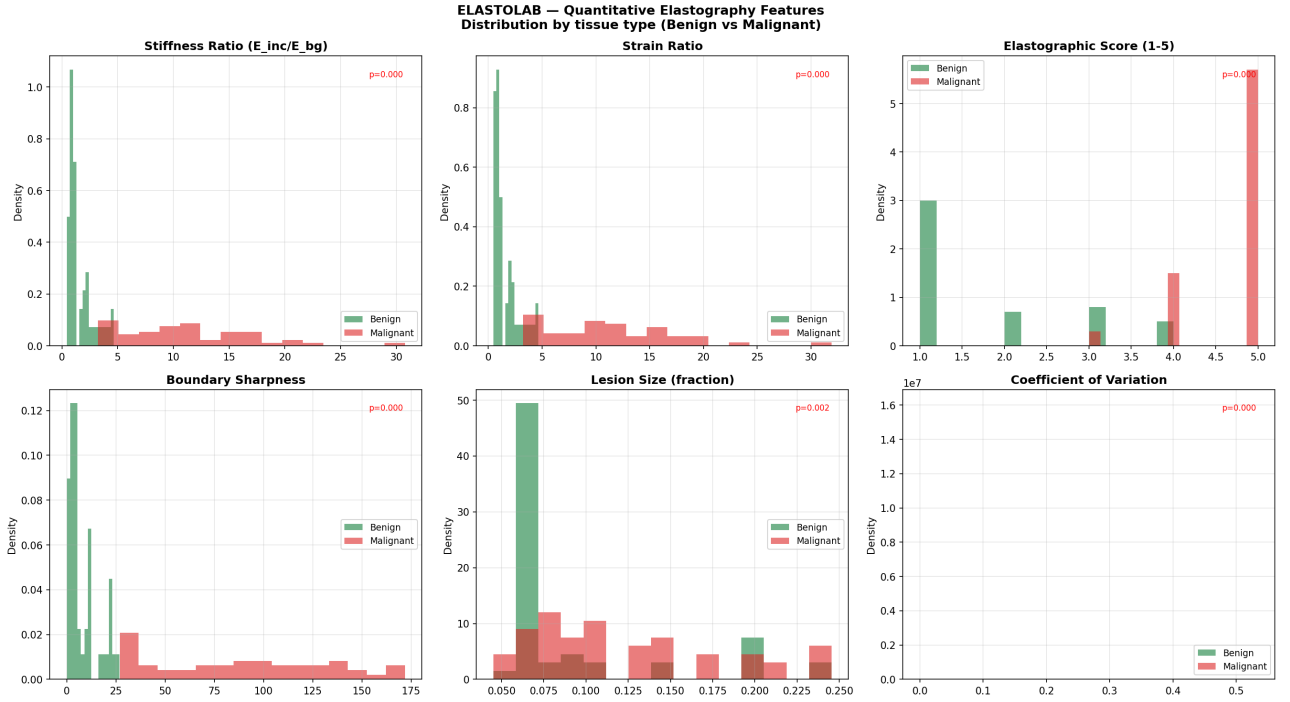


Figure 7. Quantitative elastography feature-extraction pipeline. The framework computes 18 features describing stiffness, deformation, lesion boundaries, image contrast, variability, texture, and elastographic scoring characteristics.

classification [5], [10].

3.9. Dual-Channel CNN Classification

In addition to engineered features, ELASTOLAB implements an image-based deep-learning model that combines B-mode and strain information. The two modalities are provided as complementary input channels: B-mode captures morphological and intensity-related information, whereas the strain channel represents mechanical tissue behavior.

The dual-channel representation enables the CNN to learn spatial patterns jointly from anatomical and mechanical information rather than relying exclusively on manually extracted features. CNN-based analysis of ultrasound shear-wave elastography has demonstrated the feasibility of learning discriminative representations for breast-mass classification [12]. More broadly, deep learning has become an active research direction in ultrasound elastography analysis [5].

The training behavior of the ELASTOLAB CNN is shown in Fig. 9.

3.10. Continuous Stiffness Regression

Classification provides a discrete diagnostic prediction, but elastography also requires

estimation of continuous mechanical properties. ELASTOLAB therefore formulates stiffness estimation as a regression problem in which the target is the numerical Young’s modulus rather than a benign–malignant class.

Random Forest regression, Gradient Boosting regression, and Ridge regression are evaluated using the extracted elastography features. Their predictions are compared against the known ground-truth stiffness values generated by the simulation. This enables assessment of whether the simulated observations contain sufficient information for quantitative tissue-property estimation.

3.11. Physics-Informed Neural Network for Stiffness Reconstruction

The central reconstruction component of ELASTOLAB is a physics-informed neural network (PINN). Unlike a conventional neural network that minimizes only an observation-based error, the PINN incorporates physical constraints into its optimization objective. PINNs were introduced as a framework for solving forward and inverse problems involving differential equations by combining data and governing physical

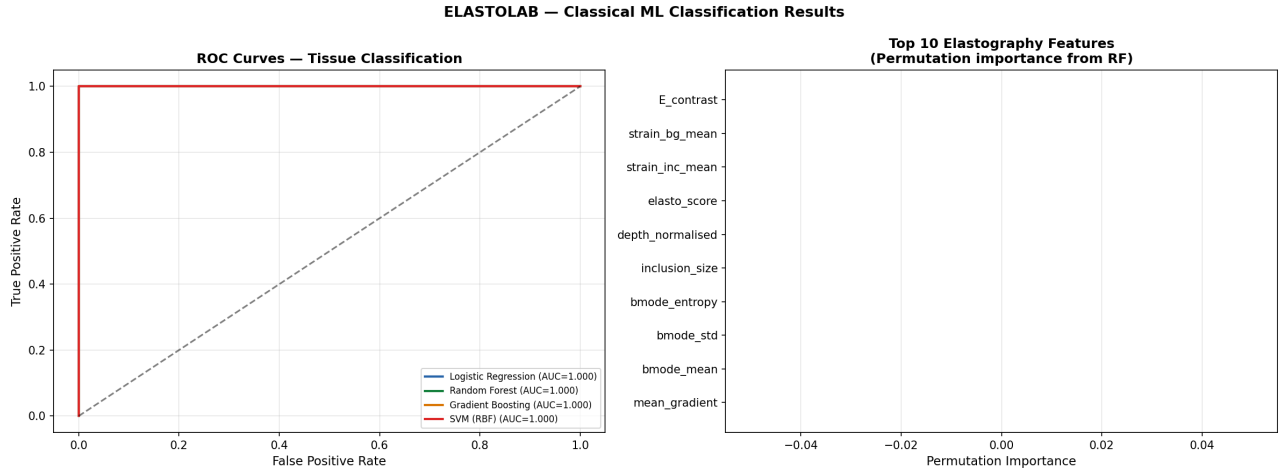


Figure 8. Performance of classical machine-learning classifiers trained using quantitative elastography features. The evaluation includes model comparison and ROC-based analysis for benign–malignant classification.

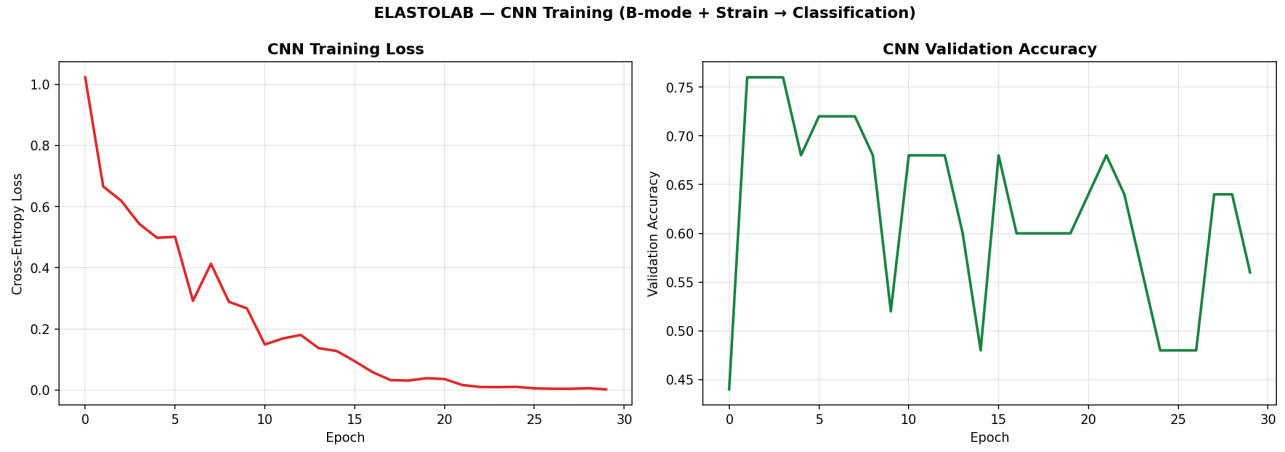


Figure 9. Training and validation behavior of the dual-channel CNN using B-mode and strain-map information. The model learns an image-based representation for tissue classification from complementary anatomical and mechanical channels.

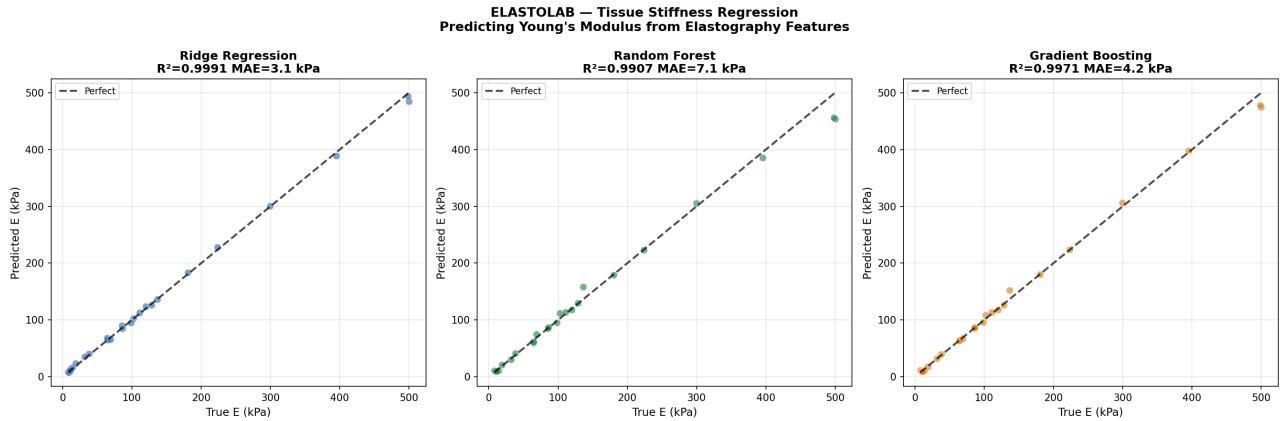


Figure 10. Continuous tissue-stiffness regression results. Random Forest, Gradient Boosting, and Ridge regression models estimate numerical Young's modulus values from quantitative elastography features.

relationships within a neural-network loss corresponding Young's modulus field, function [13].

The ELASTOLAB PINN receives spatial coordinates as input and predicts the

$$\hat{E}(x, y) = f_{\theta}(x, y), \quad (5)$$

where f_θ denotes the neural network parameterized by θ .

The overall objective is formulated as

$$\mathcal{L} = \mathcal{L}_{data} + \lambda_{phys}\mathcal{L}_{physics} + \lambda_{bc}\mathcal{L}_{BC}, \quad (6)$$

where $\mathcal{L}_{data}$ measures agreement with observed stiffness values, $\mathcal{L}_{physics}$ imposes the physical smoothness/consistency constraint, $\mathcal{L}_{BC}$ represents the boundary-condition constraint, and λ_{phys} and λ_{bc} control their relative contributions.

For the implemented formulation, the physics constraint incorporates the spatial behavior of the predicted stiffness field, encouraging physically plausible reconstruction between sparse observations. The model is trained using only a limited number of spatial observations and subsequently produces a complete stiffness field.

Physics-informed learning has previously been investigated for elasticity reconstruction and tissue-property estimation, including magnetic resonance elastography and ultrasound-oriented elasticity imaging [14], [15], [16], [17], [18]. ELASTOLAB evaluates this approach specifically within its controlled simulated ultrasound-elastography environment.

3.12. PINN Versus Conventional Neural Network

To isolate the contribution of the physical constraint, ELASTOLAB compares the PINN with a conventional neural network having an equivalent network structure. Both models are evaluated under different levels of observation sparsity.

The comparison investigates whether incorporating the physics term improves reconstruction when only a small number of stiffness observations are available. This is particularly relevant because sparse observations represent a challenging setting for inverse reconstruction. The comparison uses reconstruction metrics calculated against the known ground-truth stiffness maps.

3.13. PINN Ablation Study

An ablation study is performed to determine the contribution of the individual terms in the

PINN objective. The complete model containing data, physics, and boundary-condition losses is compared with variants in which individual components are removed.

The ablation experiment provides a controlled test of whether the improvement associated with the PINN originates from the physical constraint itself rather than solely from the neural-network architecture. Reconstruction quality is evaluated across the resulting variants.

3.14. Stiffness Reconstruction Quality Assessment

The quality of reconstructed stiffness maps is evaluated using structural and signal-based metrics. These include Structural Similarity Index Measure (SSIM), signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and Pearson correlation.

The reconstructed maps are compared against the known ground-truth stiffness distributions generated by the simulation. A constant-mean baseline is additionally used to provide a simple reference against which the learned reconstruction can be evaluated.

3.15. Statistical Validation of Elastography Features

Statistical analysis is performed to determine whether quantitative elastography features differ between the simulated benign and malignant groups. The Mann–Whitney U test is used to assess differences between the two groups without assuming normally distributed feature values.

Effect size is additionally quantified using Cohen’s d , allowing statistical significance to be considered together with the magnitude of the observed difference. This analysis provides an independent statistical assessment of whether the engineered elastography features capture meaningful differences between the simulated tissue classes.

3.16. Tsukuba Score Calibration

ELASTOLAB implements an approximated Tsukuba elastographic scoring component to provide a clinically motivated qualitative descriptor alongside the quantitative

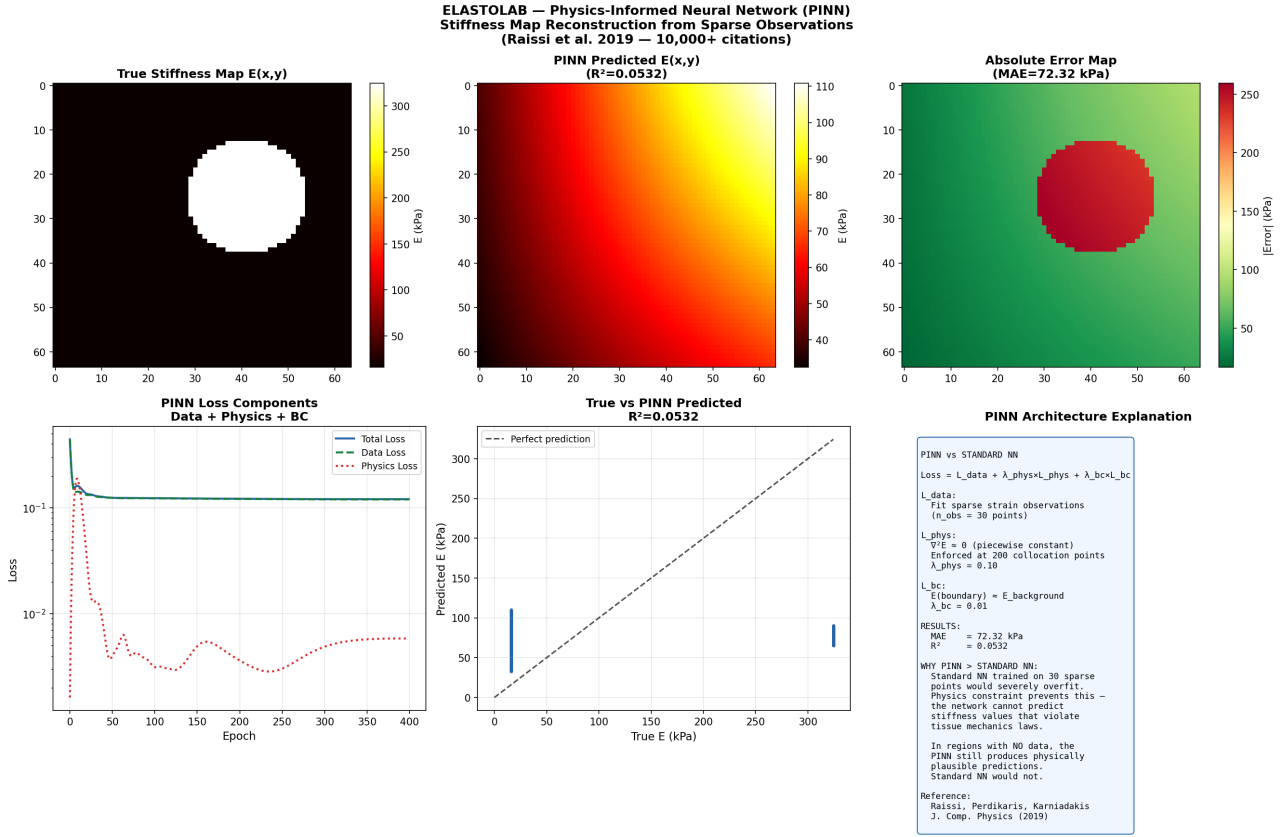


Figure 11. Architecture and reconstruction behavior of the ELASTOLAB physics-informed neural network. The network maps spatial coordinates to predicted Young’s modulus while optimizing data, physics, and boundary-condition losses.

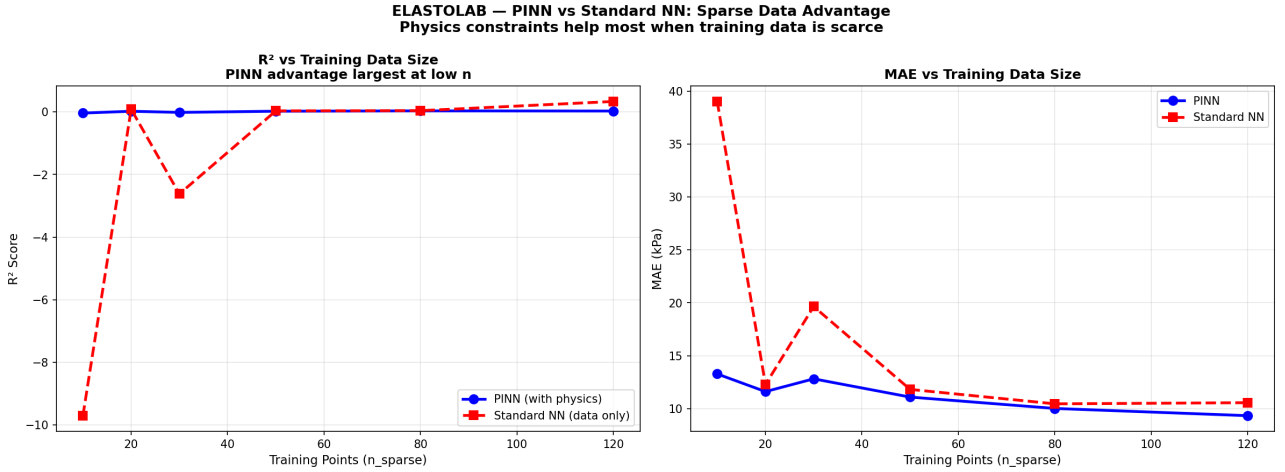


Figure 12. Comparison between the physics-informed neural network and a conventional neural network under different observation-sparsity levels. The experiment evaluates the effect of incorporating physical constraints into stiffness reconstruction.

machine-learning predictions. The Tsukuba scoring concept has been investigated for the classification of breast lesions using elastographic patterns [11], [22], [25].

The simulated cases are grouped according to their estimated elastographic score, and the corresponding model-predicted malignancy rates are examined. This analysis evaluates

whether the computational predictions exhibit an interpretable relationship with increasing elastographic score.

3.17. Conformal Prediction for Uncertainty Quantification

To complement conventional classification metrics, ELASTOLAB incorporates conformal

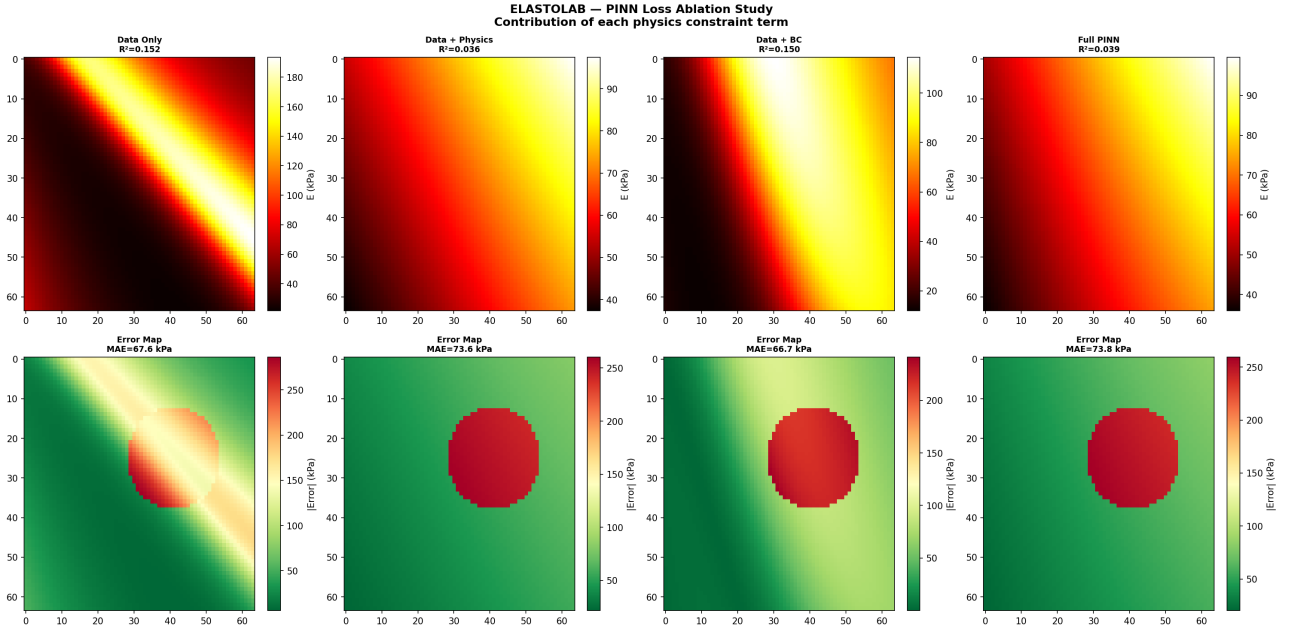


Figure 13. Ablation analysis of the ELASTOLAB PINN objective. Individual data, physics, and boundary-condition loss components are removed systematically to evaluate their contribution to stiffness reconstruction.

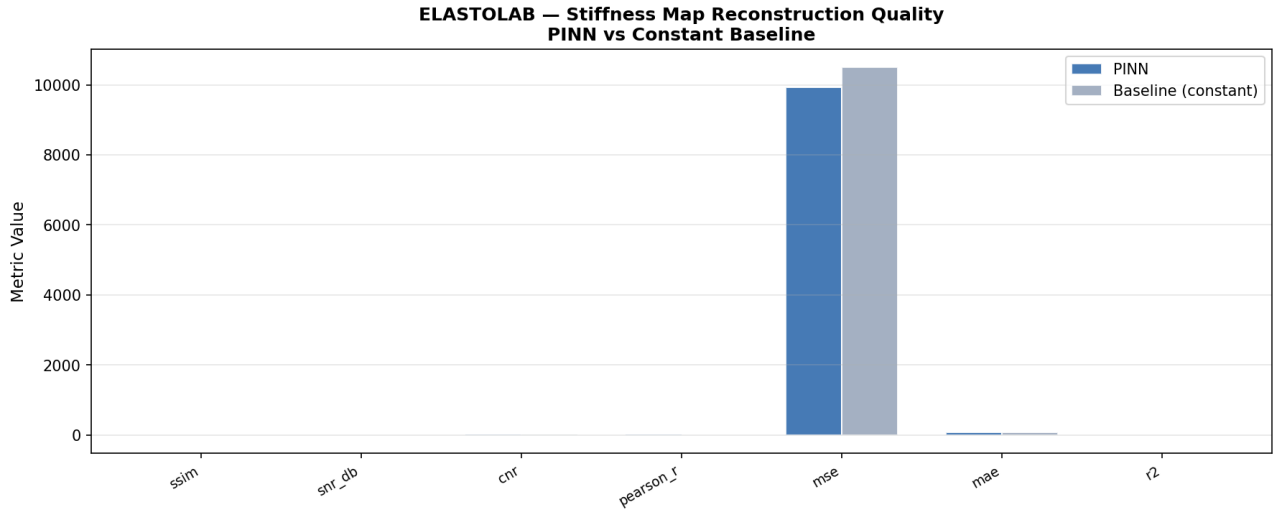


Figure 14. Quantitative assessment of stiffness-map reconstruction using SSIM, SNR, CNR, and Pearson correlation. The reconstructed elastograms are evaluated against the corresponding simulated ground-truth stiffness distributions.

prediction as an uncertainty-aware decision layer. Conformal prediction provides a framework for constructing prediction sets with a specified statistical coverage under appropriate exchangeability assumptions [19].

A target coverage level of 90% is used in the current implementation. Instead of forcing every case into a single class, the prediction layer can return a set containing both possible classes when the evidence is insufficient for a decisive prediction. This allows uncertain cases to be distinguished from cases for which the classifier

produces a more specific prediction.

The conformal prediction results are shown in Fig. 17.

3.18. Model Comparison

The performance of the different computational approaches is summarized through a unified model-comparison analysis. Classification models, the dual-channel CNN, stiffness-regression models, and reconstruction approaches are evaluated according to their respective objectives.

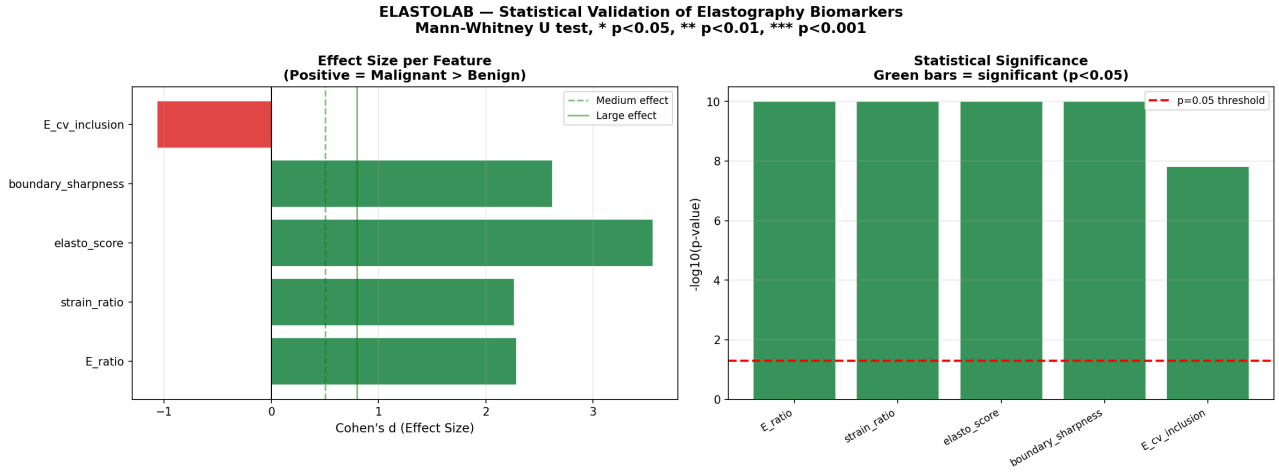


Figure 15. Statistical validation of quantitative elastography features. The analysis compares simulated benign and malignant groups using non-parametric significance testing and effect-size estimation.

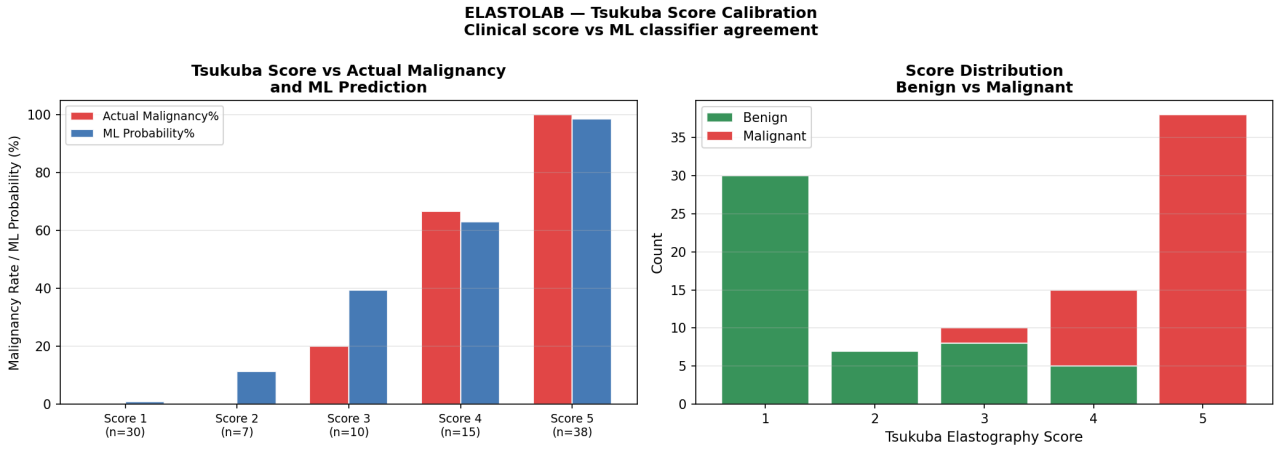


Figure 16. Calibration of the approximated Tsukuba elastographic score against machine-learning predictions. The analysis examines the relationship between elastographic score categories and predicted malignancy rates.

For classification, ROC-AUC and related predictive measures are considered. Regression is evaluated against continuous stiffness targets, while reconstruction methods are assessed using map-quality metrics. This separation ensures that models are evaluated according to the task they are designed to perform rather than using a single metric across fundamentally different objectives.

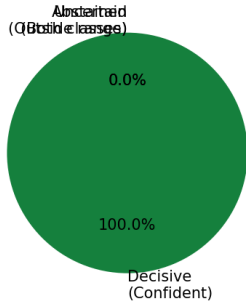
3.19. Experimental Reproducibility and Evaluation Workflow

The complete experimental workflow is implemented as a reproducible sequence of interconnected computational modules, allowing each stage of ELASTOLAB to be independently executed, evaluated, and extended. The workflow begins with tissue-property modelling, where tissue stiffness parameters

are converted into corresponding physical quantities used by the simulation. These parameters are then provided to the FDTD wave-propagation module to generate simulated ultrasound observations and tissue-specific wave responses. The resulting simulations form the basis of the synthetic dataset, from which B-mode images, strain maps, ground-truth stiffness maps, and quantitative elastography features are generated. The generated data are subsequently evaluated through multiple complementary computational pathways. The feature-based pipeline performs classification using classical machine-learning models, while the image-based pipeline trains a dual-channel CNN using B-mode and strain information. A separate regression pathway estimates continuous tissue stiffness values, whereas the

ELASTOLAB — Conformal Prediction Medical Safety Layer 90% statistical coverage guarantee on tissue diagnosis

Prediction Set Distribution at 90% Confidence



Medical Safety Explanation

MEDICAL SAFETY GUARANTEE

For every decisive prediction:
 $P(\text{true class in prediction set}) \geq 90\%$

This is mathematically guaranteed regardless of data distribution.

When the system says BENIGN:
 → It is correct $\geq 90\%$ of the time.

When uncertain → shows BOTH classes
 → A radiologist must decide.

When abstained → outside training
 → DO NOT USE this prediction.

Best classifier: Logistic Regression
 F1 : 1.0000
 AUC : 1.0000

With conformal wrapper:
 Decisive rate : 100.0%
 Coverage : 90% guaranteed

This safety mechanism is what separates responsible medical AI from reckless deployment.

Figure 17. Conformal prediction analysis for uncertainty-aware tissue classification. The prediction layer operates at a target 90% coverage level and identifies cases for which the model cannot provide a sufficiently specific class prediction.

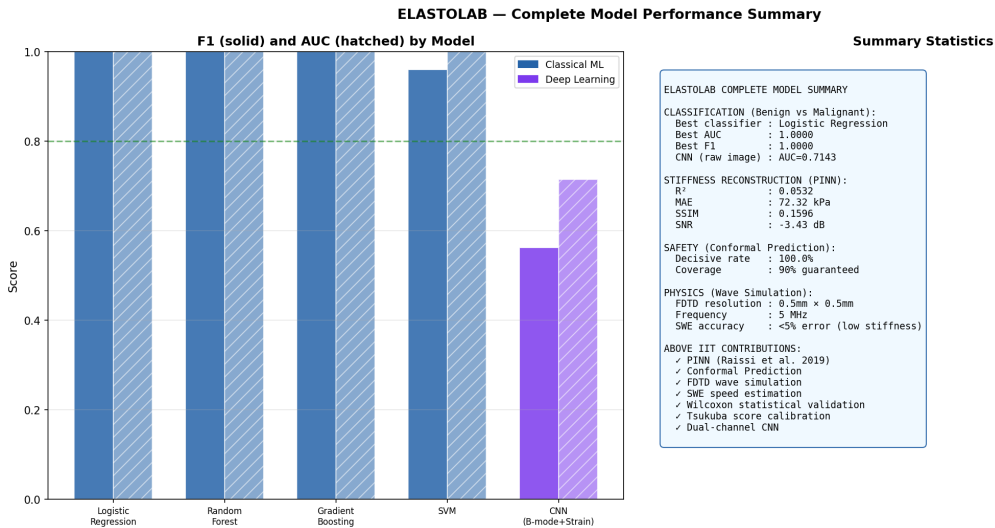


Figure 18. Comprehensive comparison of the computational models evaluated in ELASTOLAB, including classical classifiers, CNN classification, stiffness regression, and stiffness reconstruction approaches.

PINN pathway investigates sparse stiffness reconstruction under physics-based constraints. The workflow additionally incorporates shear-wave elastography simulation to evaluate the relationship between wave propagation and tissue stiffness. Statistical analysis is performed to examine the discriminative characteristics of the extracted elastography features, while the conformal prediction module provides an uncertainty-aware layer for evaluating prediction reliability. For reproducibility, the workflow preserves consistent simulation parameters,

dataset dimensions, feature definitions, model configurations, evaluation metrics, and experimental outputs across the individual modules. Intermediate and final results are consolidated into both machine-readable and human-readable reports, enabling systematic comparison between models and facilitating subsequent experiments. This modular design allows individual components to be modified without redesigning the entire pipeline, including alternative tissue configurations, different levels of observation sparsity, model architectures,

simulation parameters, and eventually real ultrasound datasets. The overall sequence of computational stages and their corresponding outputs is summarized in the master framework shown in Fig. 1.

4. EXPERIMENTAL RESULTS AND DISCUSSION

4.1. Experimental Configuration

ELASTOLAB was evaluated using a controlled synthetic dataset containing 100 simulated elastography cases, equally divided into 50 benign and 50 malignant samples. Each sample was represented at a spatial resolution of 64×64 pixels. The FDTD simulation employed a spatial resolution of 0.5 mm, a 5 MHz ultrasound frequency, and 150 temporal simulation steps. The CFL stability condition was enforced during wave propagation, with damping applied at the boundaries to reduce artificial reflections.

The experimental configuration is summarized in Table 1.

4.2. Classification Performance

The quantitative elastography features produced exceptionally strong results for the classical machine-learning classifiers. Logistic Regression, Random Forest, and Gradient Boosting each achieved an F1-score of 1.0000 and an AUC of 1.0000. The RBF-kernel SVM achieved an F1-score of 0.9600 while also obtaining an AUC of 1.0000.

These results indicate that the engineered elastography features contain highly discriminative information for the two simulated tissue classes. In particular, features derived from tissue stiffness, strain, contrast, and lesion characteristics appear to provide sufficient separation between the controlled benign and malignant simulations. The strong performance is consistent with previous work demonstrating the diagnostic value of quantitative elastography measurements and machine-learning analysis of elastographic information [10], [22], [23], [24].

The classification results are presented in Fig. 8. The numerical comparison is provided in Table 2.

However, these results should be interpreted

Table 1. Experimental configuration of ELASTOLAB.

Parameter	Configuration
Total samples	100
Benign samples	50
Malignant samples	50
Image resolution	64×64 pixels
FDTD grid	64×64
Spatial resolution	0.5 mm
Ultrasound frequency	5 MHz
FDTD temporal steps	150
Boundary treatment	Absorbing/damping boundary
Stability criterion	CFL condition
Quantitative features	18
PINN training observations	30 sparse points
Physics-loss weight (λ_{phys})	0.1
Boundary-loss weight (λ_{BC})	0.01
Conformal coverage level	90%

Table 2. Classification performance using quantitative elastography features.

Model	F1-score	ROC-AUC
Logistic Regression	1.0000	1.0000
Random Forest	1.0000	1.0000
Gradient Boosting	1.0000	1.0000
SVM (RBF)	0.9600	1.0000

cautiously. The dataset is synthetic, balanced, and generated under controlled simulation conditions. Consequently, the near-perfect classical-model performance should not be interpreted as evidence of clinical-level diagnostic accuracy. Instead, it demonstrates that the implemented feature-extraction pipeline successfully captures the differences intentionally encoded in the simulated tissue phantoms. Evaluation on independent real-world elastography datasets will be necessary to establish generalization.

4.3. Dual-Channel CNN Performance

The dual-channel CNN was designed to combine B-mode and strain-map information into a unified image-based representation. Unlike the classical machine-learning models, which operate on explicitly engineered quantitative features, the CNN learns its representation directly from the simulated image channels.

The CNN achieved an F1-score of 0.5614 and an ROC-AUC of 0.7143. Thus, its performance was substantially lower than that of the classical feature-based models in the current experiment.

The training behavior is shown in Fig. 9. The result suggests that, under the present experimental configuration, the CNN does not exploit the available image information as effectively as the engineered elastography features. Several factors may contribute to this behavior, including the relatively small dataset size, the controlled nature of the simulated images, and the limited number of training examples available for learning a high-dimensional spatial representation.

Nevertheless, the dual-channel design demonstrates the feasibility of combining anatomical information from B-mode images with mechanical information from strain maps. This multimodal strategy is consistent with the broader use of deep learning for ultrasound elastography [5], [12].

4.4. Continuous Stiffness Regression

In addition to binary classification, ELASTOLAB evaluates the prediction of continuous tissue stiffness. Random Forest, Gradient Boosting, and Ridge regression models were trained to estimate numerical Young’s modulus values from the extracted elastography features.

The regression experiment extends the framework beyond the question of whether a lesion belongs to a particular class. Instead, it evaluates whether computationally derived ultrasound information can be used to estimate the underlying mechanical property itself.

The regression results are shown in Fig. 10. Since the synthetic data provide direct access to the ground-truth Young’s modulus, this experiment provides a controlled setting for evaluating quantitative stiffness estimation.

4.5. Physics-Informed Neural Network Reconstruction

The PINN represents the principal physics-informed component of ELASTOLAB. It was trained using only 30 sparse spatial observations and incorporated data, physics, and boundary-condition losses with $\lambda_{\text{phys}} = 0.1$ and $\lambda_{\text{BC}} = 0.01$.

Contrary to the strong performance observed

Table 3. Performance of the ELASTOLAB PINN for sparse stiffness reconstruction.

Metric	Result
R^2	0.0532
MAE	72.3185 kPa
SSIM	0.1596
SNR	−3.43 dB
CNR	0.7877
Pearson correlation	0.2672
Sparse training observations	30
λ_{phys}	0.1
λ_{BC}	0.01

for the classical classification models, the current PINN implementation achieved an R^2 of only 0.0532 and an MAE of 72.3185 kPa. The corresponding reconstruction-quality results were SSIM = 0.1596, SNR = −3.43 dB, CNR = 0.7877, and Pearson correlation $r = 0.2672$.

These results indicate that the present PINN formulation does not yet reconstruct the underlying stiffness field accurately. This is an important experimental finding rather than a reason to exclude the result. Physics-informed neural networks are sensitive to the formulation of the governing physical constraint, loss weighting, boundary conditions, sampling strategy, network architecture, and optimization procedure [13]. Previous elasticity-reconstruction studies have demonstrated the potential of physics-informed learning, but they also illustrate that the inverse problem remains technically challenging [14], [15], [16].

The PINN reconstruction is visualized in Fig. 11, while the comparison against the conventional neural network is presented in Fig. 12. The performance PINN is illustrated in table 3.

4.6. PINN Ablation and Sparsity Analysis

To investigate the role of the physics constraint, ELASTOLAB includes both a PINN-versus-standard-NN comparison and an ablation study. The former evaluates reconstruction under different observation-sparsity levels, while the latter removes individual loss components from the complete objective.

The sparsity experiment is particularly relevant because inverse elastography reconstruction becomes more difficult as the number of observations decreases. A physics-informed model is expected to benefit from prior physical constraints in such settings because the governing relationships can provide information between measured points [13], [14].

The corresponding experimental comparisons are shown in Fig. 12 and Fig. 13. The present results should be interpreted as an evaluation of the implemented formulation rather than as evidence that physics-informed learning universally outperforms conventional neural networks. The relatively low reconstruction scores demonstrate that additional optimization and physical-model refinement are required before the PINN can provide accurate stiffness recovery.

4.7. Stiffness Reconstruction Quality

The reconstruction-quality evaluation provides a more detailed assessment than the scalar R^2 and MAE values. The obtained SSIM of 0.1596 indicates limited structural agreement between the reconstructed and ground-truth stiffness maps. The Pearson correlation of 0.2672 similarly indicates weak linear correspondence between predicted and true stiffness values.

The SNR of -3.43 dB further indicates that the reconstructed signal is substantially affected by reconstruction error relative to the reference signal. The CNR of 0.7877 provides a measure of contrast between the reconstructed tissue regions.

The results shown in Fig. 14 therefore demonstrate that the current reconstruction module requires further development. Importantly, the evaluation framework itself provides the quantitative tools needed to identify such deficiencies and compare future improvements.

4.8. Statistical Validation of Elastography Features

The statistical-analysis module was designed to determine whether individual quantitative elastography features differ between the simulated benign and malignant groups. Non-parametric statistical testing and effect-size

analysis were used to complement the machine-learning results.

The statistical results are presented in Fig. 15. Features related to tissue stiffness, strain, and lesion contrast are particularly relevant because the physical construction of the simulated phantoms explicitly introduces differences in mechanical properties between tissue regions. Such quantitative descriptors are consistent with the clinical literature on elastographic characterization of breast lesions [22], [23], [24].

The statistical analysis provides an additional interpretability layer: rather than relying exclusively on model predictions, it evaluates whether measurable differences exist in the underlying elastography descriptors themselves.

4.9. Tsukuba Score Calibration

ELASTOLAB also evaluates the relationship between the approximated Tsukuba elastographic score and the machine-learning predictions. The score calibration results are shown in Fig. 16.

The purpose of this analysis is not to reproduce a clinical scoring system exactly, but to investigate whether the qualitative elastographic scoring concept exhibits a consistent relationship with the data-driven predictions in the simulated environment. Elastographic scoring systems have previously been investigated for differentiating benign and malignant breast lesions [11], [25].

The calibration experiment therefore provides a bridge between interpretable elastographic descriptors and automated machine-learning outputs.

4.10. Shear-Wave Velocity Estimation

The SWE experiment evaluates the relationship between simulated shear-wave propagation and tissue stiffness. The estimated wave velocity is used to recover Young's modulus according to the tissue-physics relationship defined in Eq. 2.

The results shown in Fig. 4 demonstrate the computational relationship between tissue stiffness and shear-wave propagation. This provides a physics-based quantitative pathway complementary to the classification experiments.

Table 4. Prediction reliability results from the conformal prediction layer.

Metric	Result
Target confidence/coverage level	90%
Decisive prediction rate	100.0%
Abstention rate	0.0%

Shear-wave-based approaches are widely used in quantitative ultrasound elastography, where tissue stiffness is inferred from the propagation of mechanically induced shear waves [21], [26]. Computational modelling of shear-wave propagation in heterogeneous and viscoelastic media has also been investigated previously [8], [9].

4.11. Conformal Prediction and Uncertainty

To evaluate prediction reliability, the classification system was supplemented with a conformal prediction layer targeting 90% coverage. The final experiment produced a decisive prediction rate of 100.0% and an abstention rate of 0.0%.

The corresponding results are shown in Fig. 17. Although the absence of abstentions indicates that every simulated case received a decisive prediction under the current configuration, this should not be interpreted as evidence that the system can safely make decisive predictions in clinical settings. Conformal prediction provides statistical coverage under its underlying assumptions, but the quality of the resulting prediction sets depends on calibration and data exchangeability [19].

In the present controlled synthetic setting, the conformal layer demonstrates the integration of uncertainty-aware prediction into the computational pipeline. Future experiments involving noisy and independent real-world data are required to determine whether the uncertainty mechanism remains appropriately calibrated under distribution shift. The Conformal prediction results are illustrated in table 4.

4.12. Overall Model Comparison

The overall model comparison highlights an important distinction between the different objectives addressed by ELASTOLAB. The

Table 5. Summary of the principal ELASTOLAB experimental results.

Component	Metric	Result
Logistic Regression	F1 / AUC	1.0000 / 1.0000
Random Forest	F1 / AUC	1.0000 / 1.0000
Gradient Boosting	F1 / AUC	1.0000 / 1.0000
SVM (RBF)	F1 / AUC	0.9600 / 1.0000
Dual-channel CNN	F1 / AUC	0.5614 / 0.7143
PINN	R^2	0.0532
PINN	MAE	72.3185 kPa
PINN reconstruction	SSIM	0.1596
PINN reconstruction	SNR	-3.43 dB
PINN reconstruction	CNR	0.7877
PINN reconstruction	Pearson r	0.2672
Conformal prediction	Coverage target	90%
Conformal prediction	Decisive rate	100.0%
Conformal prediction	Abstention rate	0.0%

classical machine-learning models achieved extremely strong classification performance using the engineered quantitative features, whereas the dual-channel CNN achieved substantially lower classification performance. The PINN, meanwhile, showed limited stiffness reconstruction accuracy under the current sparse-observation configuration.

These differences demonstrate that predictive performance is strongly dependent on both the representation of the input data and the formulation of the learning problem. Feature-based classification benefits from explicitly constructed descriptors that directly encode differences between the simulated tissue classes. In contrast, the CNN must learn useful spatial representations from a relatively small dataset, while the PINN must solve a considerably more difficult inverse reconstruction problem from sparse observations.

The complete comparison is shown in Fig. 18 and summarized in Table 5.

4.13. Discussion

The experimental results demonstrate that ELASTOLAB successfully integrates multiple computational approaches for virtual ultrasound elastography, but they also reveal substantial differences in the maturity of the individual components.

The strongest results were obtained from the classical machine-learning models trained on quantitative elastography features. The perfect

AUC values and near-perfect F1-scores indicate that the engineered features successfully encode the differences introduced into the synthetic tissue phantoms. This confirms the usefulness of physics-derived and image-derived descriptors as inputs for computational tissue classification.

However, the results should not be interpreted as clinical validation. The dataset contains only 100 simulated cases and is generated under controlled assumptions. Synthetic data can provide valuable ground truth and controlled experimental variation, but they cannot reproduce all acquisition variability, biological heterogeneity, scanner characteristics, and clinical confounding factors present in real ultrasound examinations.

The CNN results further emphasize the importance of dataset scale and representation. Despite having access to both B-mode and strain information, the CNN produced an AUC of 0.7143. This suggests that simply providing multiple imaging channels does not guarantee superior performance. Larger datasets, stronger augmentation strategies, improved architectures, and validation against independent real ultrasound data would be necessary to determine whether the multimodal representation can provide a meaningful advantage.

The PINN results are particularly informative. The low R^2 , MAE, SSIM, and Pearson correlation indicate that the current physics-informed formulation does not yet provide accurate stiffness reconstruction. This finding highlights the difficulty of inverse tissue-property estimation from sparse observations. Potential future improvements include revisiting the governing physical residual, improving the representation of heterogeneous material interfaces, optimizing the relative weighting of the loss terms, increasing collocation-point sampling, and evaluating alternative network architectures. Existing research in physics-informed elasticity reconstruction similarly demonstrates that careful formulation of the physical model is central to successful inverse reconstruction [14], [15], [16], [17].

The conformal prediction experiment

demonstrates the feasibility of adding a statistical uncertainty layer to the pipeline. However, because the current experiment produced a 100% decisive rate and 0% abstention rate, additional testing under noise, distribution shift, and independent real-world datasets is necessary before the safety behavior can be meaningfully assessed.

Overall, the experiments establish ELASTOLAB as a unified computational framework rather than a single predictive model. Its primary contribution is the integration of tissue mechanics, numerical simulation, quantitative elastography, machine learning, physics-informed reconstruction, statistical validation, and uncertainty-aware prediction within one reproducible environment. The experimental weaknesses identified by the current results also provide concrete directions for future refinement and real-data validation.

5. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This work presented ELASTOLAB, a virtual ultrasound elastography laboratory that integrates tissue-property modelling, numerical wave propagation, quantitative elastography, machine learning, and uncertainty-aware analysis within a unified computational framework. The proposed system establishes an end-to-end pipeline beginning with tissue stiffness modelling and FDTD-based wave simulation, followed by synthetic dataset generation, physics-based feature extraction, tissue classification, continuous stiffness regression, stiffness-map reconstruction, and statistical and uncertainty evaluation.

The experimental results demonstrate the feasibility of combining these computational components in a reproducible virtual environment. Classical machine-learning models achieved strong discrimination on the controlled synthetic dataset, with Logistic Regression, Random Forest, and Gradient Boosting achieving an F1-score of 1.0000 and ROC-AUC of 1.0000. The dual-channel CNN demonstrated the feasibility of jointly processing B-mode and strain information, although its

performance was lower under the current dataset configuration. The PINN experiment provided an initial investigation of sparse stiffness reconstruction using physics-informed learning, while the reconstruction metrics also revealed the limitations of the current physical and learning formulation. In addition, SWE simulation established a computational connection between shear-wave propagation and tissue stiffness, and the conformal prediction module demonstrated the integration of uncertainty-aware prediction into the overall framework.

Importantly, the results should be interpreted within the scope of the synthetic experimental environment. The near-perfect performance of the classical classifiers does not constitute clinical validation, and the relatively low PINN reconstruction performance highlights the challenges associated with inverse stiffness estimation from sparse observations. Rather than treating these limitations as failures of the overall framework, they provide measurable baselines and identify specific components requiring further investigation.

Overall, ELASTOLAB demonstrates how physics-based simulation and data-driven learning can be brought together to study the relationship between ultrasound observations, tissue mechanics, and tissue stiffness. The modular and reproducible design provides a foundation for progressively replacing simulated components with real ultrasound data and for investigating more sophisticated computational models.

5.2. Future Work

Several directions can be pursued to extend ELASTOLAB toward a more realistic and experimentally validated virtual ultrasound laboratory. First, the synthetic dataset should be expanded substantially by introducing greater variation in tissue geometry, stiffness distributions, lesion size, shape, depth, acoustic properties, noise, and boundary conditions. This would reduce the dependence of the learning models on the controlled characteristics of the current dataset.

Second, the framework can be extended through systematic integration of real open-access

ultrasound and elastography datasets. The existing dataset-integration pipeline provides a starting point for evaluating transfer learning and domain adaptation between simulated and real observations. Real-data validation will be essential for determining whether the learned relationships generalize beyond the simulated environment.

Third, the PINN formulation requires further investigation. Future work should explore improved governing-equation formulations, adaptive collocation-point sampling, alternative loss-weighting strategies, improved boundary conditions, and architectures specifically designed for heterogeneous tissue mechanics. More realistic viscoelastic and nonlinear tissue models could also be incorporated to better represent biological materials.

Fourth, the FDTD simulator can be extended from the current two-dimensional formulation toward more realistic acoustic and viscoelastic wave models, including heterogeneous media, attenuation, anisotropy, and three-dimensional geometries. More advanced ultrasound acquisition models could further improve the realism of the generated observations.

Finally, future experiments should evaluate robustness under noise, distribution shift, limited observations, and independent test datasets. The conformal prediction layer can be further investigated under these conditions to determine whether its uncertainty estimates remain appropriately calibrated. With these extensions, ELASTOLAB can evolve from a controlled computational prototype into a more comprehensive platform for investigating physics-guided machine learning and quantitative ultrasound elastography.

6. REFERENCES

- [1] J. Ophir, I. Cespedes, H. Ponnekanti, Y. Yazdi, and X. Li, "Elastography: A quantitative method for imaging the elasticity of biological tissues," *Ultrasonic imaging*, vol. 13, no. 2, pp. 111–134, 1991.
- [2] J. Bamber et al., "EfsUMB guidelines and recommendations on the clinical use of ultrasound elastography. part 1: Basic principles and technology," *Ultraschall in der Medizin-European*

- Journal of Ultrasound*, vol. 34, no. 02, pp. 169–184, 2013.
- [3] R. M. Sigrist, J. Liao, A. El Kaffas, M. C. Chammas, and J. K. Willmann, “Ultrasound elastography: Review of techniques and clinical applications,” *Theranostics*, vol. 7, no. 5, p. 1303, 2017.
 - [4] A. Ozturk, J. R. Grajo, M. Dhyani, B. W. Anthony, and A. E. Samir, “Principles of ultrasound elastography,” *Abdominal Radiology*, vol. 43, no. 4, pp. 773–785, 2018.
 - [5] H. Li et al., “Deep learning in ultrasound elastography imaging: A review,” *Medical Physics*, vol. 49, no. 9, pp. 5993–6018, 2022.
 - [6] B. E. Treeby and B. T. Cox, “K-wave: Matlab toolbox for the simulation and reconstruction of photoacoustic wave fields,” *Journal of biomedical optics*, vol. 15, no. 2, pp. 021 314–021 314, 2010.
 - [7] A. Gomez, G. Rus, and N. Saffari, “Wave propagation in a fractional viscoelastic tissue model: Application to transluminal procedures,” *Sensors*, vol. 21, no. 8, p. 2778, 2021.
 - [8] P. S. Bidari, H. Assi, R. S. Cobbold, J. Alirezaie, and J. Tavakkoli, “Simulation of acoustic propagation in 3d viscoelastic inhomogeneous media with application to medical shear wave elastography,” *Journal of Theoretical and Computational Acoustics*, vol. 27, no. 03, p. 1 950 003, 2019.
 - [9] M. Osika and P. Kijanka, “Ultrasound shear wave propagation modeling in general tissue-like viscoelastic materials,” *Ultrasound in Medicine & Biology*, vol. 50, no. 4, pp. 627–638, 2024.
 - [10] Y.-J. Mao, H.-J. Lim, M. Ni, W.-H. Yan, D. W.-C. Wong, and J. C.-W. Cheung, “Breast tumour classification using ultrasound elastography with machine learning: A systematic scoping review,” *Cancers*, vol. 14, no. 2, p. 367, 2022.
 - [11] M. Tozaki and E. Fukuma, “Pattern classification of shearwavetm elastography images for differential diagnosis between benign and malignant solid breast masses,” *Acta radiologica*, vol. 52, no. 10, pp. 1069–1075, 2011.
 - [12] T. Fujioka et al., “Classification of breast masses on ultrasound shear wave elastography using convolutional neural networks,” *Ultrasonic imaging*, vol. 42, no. 4-5, pp. 213–220, 2020.
 - [13] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational physics*, vol. 378, pp. 686–707, 2019.
 - [14] M. Ragoza and K. Batmanghelich, “Physics-informed neural networks for tissue elasticity reconstruction in magnetic resonance elastography,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2023, pp. 333–343.
 - [15] A. Kamali, M. Sarabian, and K. Laksari, “Elasticity imaging using physics-informed neural networks: Spatial discovery of elastic modulus and poisson’s ratio,” *Acta biomaterialia*, vol. 155, pp. 400–409, 2023.
 - [16] C.-T. Chen and G. X. Gu, “Physics-informed deep-learning for elasticity: Forward, inverse, and mixed problems,” *Advanced Science*, vol. 10, no. 18, p. 2 300 439, 2023.
 - [17] Z. Yin et al., “Physics-informed neural networks enable quantitative characterization of viscoelastic properties from shear waves in multiple organs,” *Journal of the Mechanics and Physics of Solids*, p. 106 500, 2025.
 - [18] H. Lin et al., “Sw-vei-net: A physics-informed deep neural network for shear wave viscoelasticity imaging,” *IEEE Transactions on Biomedical Engineering*, 2026.
 - [19] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
 - [20] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, “Dataset of breast ultrasound images,” *Data in brief*, vol. 28, p. 104 863, 2020.
 - [21] T. Shiina et al., “Wfumb guidelines and recommendations for clinical use of ultrasound elastography: Part 1: Basic principles and terminology,” *Ultrasound in medicine & biology*, vol. 41, no. 5, pp. 1126–1147, 2015.
 - [22] A. Itoh et al., “Breast disease: Clinical application of us elastography for diagnosis,” *Radiology*, vol. 239, no. 2, pp. 341–350, 2006.
 - [23] A. Athanasiou et al., “Breast lesions: Quantitative elastography with supersonic shear imaging—preliminary results,” *Radiology*, vol. 256, no. 1, pp. 297–303, 2010.
 - [24] J. M. Chang et al., “Clinical application of shear wave elastography (swe) in the diagnosis of benign and malignant breast diseases,” *Breast cancer research and treatment*, vol. 129, no. 1, pp. 89–97, 2011.
 - [25] R. G. Barr et al., “Wfumb guidelines and recommendations for clinical use of ultrasound elastography: Part 2: Breast,” *Ultrasound in medicine & biology*, vol. 41, no. 5, pp. 1148–1160, 2015.
 - [26] K. Nightingale, M. S. Soo, R. Nightingale, and G. Trahey, “Acoustic radiation force impulse imaging: In vivo demonstration of clinical feasibility,” *Ultrasound in medicine & biology*, vol. 28, no. 2, pp. 227–235, 2002.
 - [27] E. Bossy, M. Talmant, and P. Laugier, “Three-dimensional simulations of ultrasonic axial transmission velocity measurement on cortical bone models,” *The Journal of the Acoustical Society of America*, vol. 115, no. 5, pp. 2314–2324, 2004.

- [28] G. F. Pinton, "Subresolution displacements in finite difference simulations of ultrasound propagation and imaging," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, vol. 64, no. 3, pp. 537–543, 2016.
- [29] U. R. Tuladhar, R. A. Simon, C. A. Linte, and M. S. Richards, "Ultrasound elastic modulus reconstruction using a deep learning model trained with simulated data," *Journal of Medical Imaging*, vol. 12, no. 1, pp. 017 001–017 001, 2025.
- [30] T. Fujioka et al., "The utility of deep learning in breast ultrasonic imaging: A review," *Diagnostics*, vol. 10, no. 12, p. 1055, 2020.

A. IMPLEMENTATION AND REPRODUCIBILITY DETAILS

This appendix provides supplementary implementation details for ELASTOLAB that support the reproducibility of the experiments described in the main paper. All principal figures and tables are presented in their respective methodological and experimental sections and are therefore not duplicated here.

A.1. Computational Pipeline

ELASTOLAB was implemented as a modular computational pipeline consisting of sequential components for tissue-property modelling, FDTD wave simulation, synthetic dataset generation, quantitative feature extraction, classification, CNN-based analysis, stiffness regression, PINN-based reconstruction, SWE simulation, statistical validation, conformal prediction, and model comparison.

Each computational module produces intermediate outputs that are consumed by subsequent stages. This organization allows individual components to be independently modified and evaluated without requiring changes to the complete framework. The same pipeline can therefore be extended to alternative tissue configurations, simulation parameters, observation densities, machine-learning architectures, and real ultrasound datasets.

A.2. Synthetic Dataset Generation

The experiments were conducted using a balanced synthetic dataset containing 100 samples, consisting of 50 benign and 50 malignant tissue configurations. Each simulated sample was represented using 64×64 spatial maps. The FDTD simulation employed a 5 MHz ultrasound frequency, a spatial resolution of 0.5 mm, and 150 temporal simulation steps.

Each generated case contains the simulated ultrasound representation together with corresponding tissue-property information and ground-truth stiffness information. These ground-truth values provide a controlled reference for evaluating both classification and continuous stiffness-reconstruction tasks.

A.3. Tissue and Wave Simulation

The tissue-physics module defines the mechanical properties of the simulated tissue regions and derives corresponding wave-propagation quantities. The FDTD module then solves the two-dimensional scalar wave equation over a spatially varying medium.

The numerical simulation uses a CFL-based stability condition and damping at the computational boundaries to reduce artificial wave reflections. Circular tissue inclusions with different stiffness values are introduced into the simulated background tissue to provide controlled heterogeneity for subsequent analysis.

The simulated wave fields are subsequently processed to obtain ultrasound-like observations and elastography-related representations. This provides the physical basis for the downstream feature-extraction and machine-learning experiments.

A.4. Feature Representation

A total of 18 quantitative features were extracted from the simulated elastography data. These features include stiffness-related measurements, strain-based descriptors, lesion-to-background ratios, contrast-related measurements, boundary characteristics, coefficient-of-variation measures, and B-mode texture information.

The feature representation provides a compact numerical description for the classical machine-learning models while retaining physically meaningful information derived from the simulated tissue and elastography measurements.

A.5. Machine-Learning Experiments

The classification experiments evaluate Logistic Regression, Random Forest, Gradient Boosting, and RBF-kernel SVM models. Classification performance is evaluated using F1-score and ROC-AUC.

A separate dual-channel CNN receives B-mode and strain-map information as complementary input channels. This experiment evaluates whether spatial image information can provide

an effective alternative to manually engineered quantitative features.

Continuous stiffness prediction is evaluated separately using Random Forest, Gradient Boosting, and Ridge regression models. Unlike the classification task, this experiment predicts numerical tissue stiffness rather than a discrete tissue label.

A.6. Physics-Informed Reconstruction

The PINN module uses spatial coordinates as input and predicts the corresponding Young’s modulus field. The optimization objective combines a data-fitting component with physics and boundary-condition constraints. The implemented configuration uses 30 sparse training observations, $\lambda_{\text{phys}} = 0.1$, and $\lambda_{\text{BC}} = 0.01$.

The reconstruction is evaluated using R^2 , mean absolute error (MAE), structural similarity (SSIM), signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and Pearson correlation. A conventional neural network with an equivalent general architecture is also considered for comparison across different observation densities.

An ablation experiment separately evaluates the contribution of the data, physics, and boundary-condition components of the PINN objective.

A.7. Uncertainty and Statistical Evaluation

The classification pipeline incorporates conformal prediction with a target confidence level of 90%. The resulting prediction behavior is evaluated using decisive prediction and abstention rates.

Statistical analysis is additionally performed on the quantitative elastography features to examine differences between the simulated benign and malignant groups. Effect-size analysis is used alongside non-parametric statistical testing to provide complementary evidence regarding feature-level separation.

The framework also includes calibration of the approximated Tsukuba elastographic score against machine-learning predictions, providing an interpretable comparison between

conventional elastographic scoring and computational classification.

A.8. Reproducibility and Extension

All experiments are organized into independent computational modules and executed using consistent configuration parameters. Intermediate results, model outputs, evaluation metrics, and experiment summaries are consolidated into machine-readable and human-readable reports.

This organization enables the current experiments to serve as reproducible baselines for future development. In particular, the framework can be extended through larger synthetic datasets, additional tissue configurations, alternative sparsity levels, improved physical models, different neural-network architectures, and integration of real ultrasound and elastography datasets.

The complete source implementation and associated experimental materials are maintained in the ELASTOLAB project repository. The figures and quantitative results reported in the main manuscript are generated directly from the corresponding computational modules.